



Bidirectional Security Gateway

Technology Overview

Terafence is proud to offer a unique, secure, and innovative network security appliance – **Terafence Bidirectional Gateway**.

Terafence manufactures Cyber Security hardware products to allow security architects the freedom in protecting their networks as needed.

Terafence **TFG121-BSG** is designed to provide a better-than-ever security by filtering LAN traffic via Terafence's proprietary FPGA-based hardware architecture.

Terafence **TFG121-BSG** allows network architects to define traffic patterns through the device with total assurance that any hacking attempts fail due to galvanic separation and super strong appliance architecture.

Terafence **TFG121-BSG** utilizes dual-proprietary FPGA-based hardware architecture data diodes by Terafence to create an appliance that is protocol agnostic and yet extremely safe and secure.

Only pre-defined traffic can pass through without allowing direct TCP/IP sessions between client and server filtered by hardware-based traffic filters.

Why choose TFG121-BSG?

Key Features:

- Complete **galvanic network isolation** for enhanced security and protection.
- Acts as **fully transparent proxy** in the network
- **Core security hardware** operates independently **without an operating system, MAC address, or IP address**.
- Functions as **Layer-2 Physical Firewall** for inline traffic enforcement
- Near Real-time **data transfer** with **minimal latency**
- Secure, Electronically-controlled data exchange
- Operates within the same network segment only
- Hardware-based Packet filtering
- Port-based Management; Protocol Agnostic

Security Features:

- Hardened Operating System on the integrated CPUs
- Secure Configuration & Management access (HTTPS) to GUI

Technical Specifications:

- **Data Throughput:** Up to 1 Gbps with auto-negotiation
- **Power Supply:** 100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
- **Form Factor:** 1U rack-mountable appliance
- **Operating Temperature:** 32°F to 113°F
- **Storage Temperature:** -40 °F to 185 °F
- **Relative Humidity:** 5% ~ 95% | 55°C, non-condensing
- **Processor:** Intel i3 or better
- **RAM** – Customizable as per requirement
- **Local Storage** – Customizable as per requirement
- **Network Interfaces:** 1 x Data Port, 1 x Management Port per side
- **Relative Humidity:** - 5% ~ 95% | 131°F, non-condensing
- **In-door use only**